Hypotheses: reality **versus** our decision

We **will never know for sure** if either Ho or Ha are true (***what happens at the population level is unknown***). Thus, we proceed as follow:

- If the data offer evidence against Ho, **we reject Ho** in favor of Ha.
- If the data do not offer evidence against Ho, **we do not reject Ho.**

- We make the decision (reject or do not reject Ho) based on sample data; thus, there is always the possibility that we could have made an erroneous decision.

	True state of nature (Unknown)	
	Ho true	Ho false
Our conclusion		
Do not reject Ho	Correct decision	Type II error
Reject Ho	Type I error	Correct decision

- The probability of committing a Type I error is denoted by α.

- The probability of committing a Type II error is denoted by **ß**.